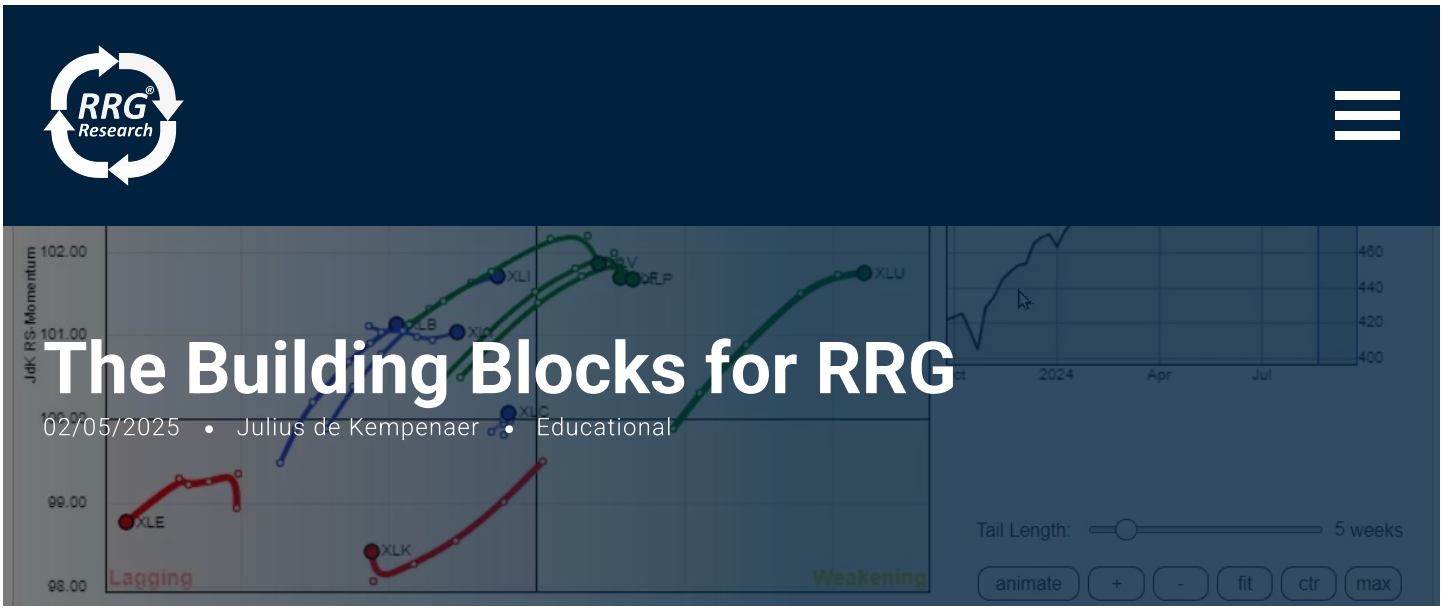


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Main Components of Relative Rotation Graphs

The two main inputs for a Relative Rotation Graph are the JdK RS-Ratio and JdK RS-Momentum.

Note that both input indicators are “normalized,” which means they are expressed in the same unit of measure and fluctuate above/below the same level (100). This normalization process means the JdK RS-Ratio values for different securities can be compared as long as the same benchmark is used.

JdK RS-Ratio

RS-Ratio is an indicator that measures the trend for relative performance. Similar to the price relative, RS-Ratio uses ratio analysis to compare one security against another (usually the benchmark). It is designed to define the trend in relative performance and measure the strength of that trend.

The chart below shows the Technology SPDR (XLK) in the main window, the price relative (XLK:$SPX ratio) in the middle window, and the RRG indicators in the bottom pane of the chart.

We will focus on RS-Ratio (red) first. RS-Momentum (green) will be covered in the next section.



Chart of Technology Select SPDR ETF (XLK), price relative (XLK:$SPX), and RRG indicators.

RS-Ratio provides a clear tool for defining the trend in relative performance. This indicator reflects an uptrend in relative performance when above 100 (relative strength) and a downtrend in relative performance when below 100 (relative weakness).

The further the indicator is above 100, the stronger the uptrend in relative performance. The further the indicator is below 100, the stronger the downtrend in relative performance.

As with all trend-following indicators, the trend-following model that powers RS-Ratio includes a lag period. This means there will already be upward movement in the price

bottomed in mid-July, but the RS-Ratio did not cross above 100 until mid-September. This is typical for trend-following indicators, which are designed to ignore the blips and focus on the trend.

The chart below shows the Consumer Discretionary SPDR (XLY) with another example.

Chart of Consumer Discretionary Select SPDR ETF (XLY), price relative (XLY:$SPX), and RRG indicators.

First, all four have RS ratios above 100, showing relative strength (against the S&P 500).

Second, XLV shows the most relative strength because its RS ratio is the highest. XLF is the weakest of the four because its RS ratio is the lowest.

JdK RS-Momentum

Before looking at RS-Momentum in detail, let's review the concept behind momentum and how it relates to trends.

As with price charts, remember that momentum changes course before the trend reverses. Not all momentum moves, however, result in trend reversals.

Consider an example using price and a moving average. Price first moves towards the moving average and then crosses it if the move extends. Price, however, doesn't always cross the moving average to signal a trend reversal. Aggressive traders would more likely take a position as the price moves toward the moving average because this means momentum is improving. Conservative investors would more likely wait for the price to move above the moving average because the trend has not fully reversed.

RS-Momentum is an indicator that measures the momentum (rate-of-change) of RS-Ratio. As a momentum indicator, it leads RS-Ratio and can be used to anticipate turns in RS-Ratio.

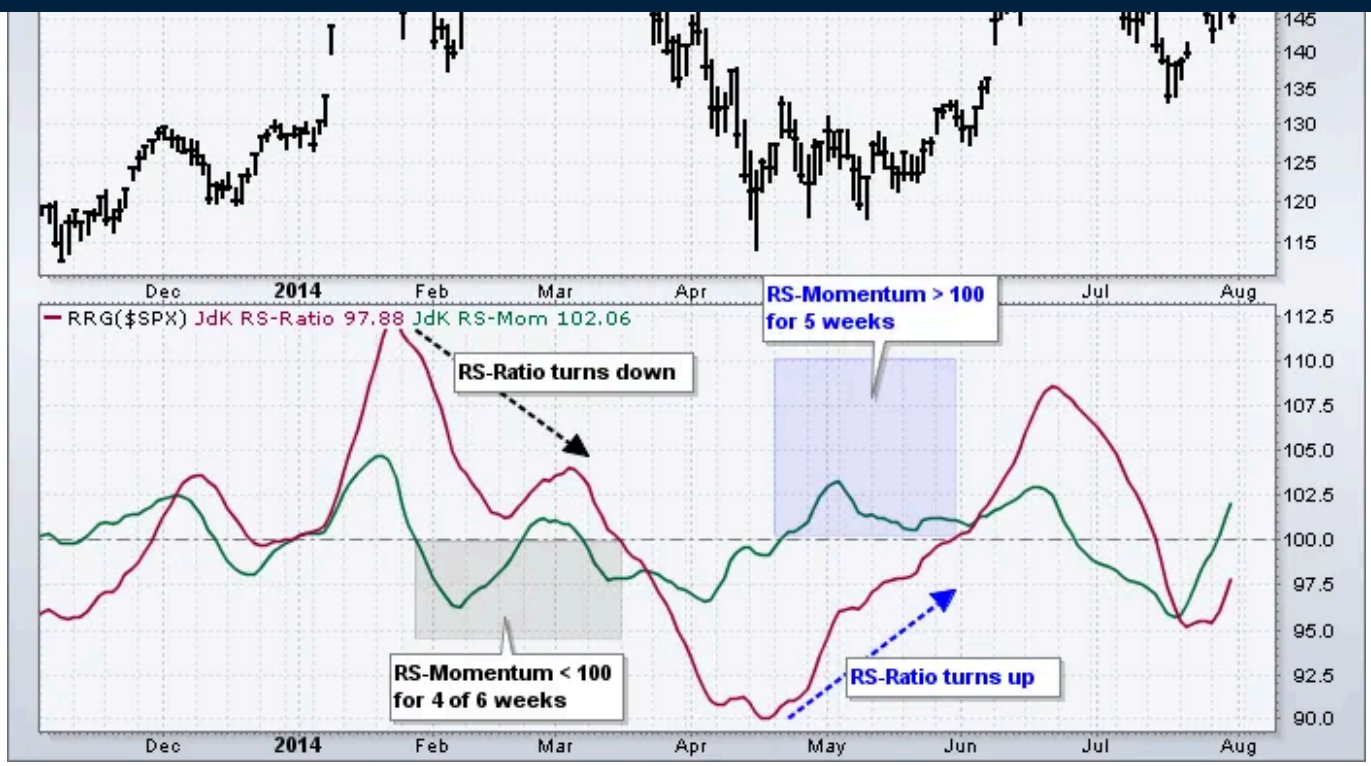
Typically, RS-Momentum crosses above 100 when RS-Ratio is forming a trough and starting to move up. Conversely, RS-Momentum crosses below 100 when RS-Ratio is forming a peak and starting to move down.

The chart below shows the Utilities SPDR (XLU) with RS-Momentum in green and RS-Ratio in red. RS-Momentum crossed above 100 in mid-December and held mostly above 100 for four weeks. Notice how RS-Ratio bottomed as RS-Momentum moved above 100 and RS-Ratio crossed above 100 later in January.

Keep in mind that RS-Momentum is an indicator of an indicator (RS-Ratio). Furthermore, as a momentum indicator, it will move often above/below the 100 level. Chartists may want to focus on sustained moves above/below 100 to anticipate a similar cross in RS-Ratio.

The chart below shows the Biotech SPDR (XBI) with two examples highlighting the relationship between RS-Momentum and RS-Ratio. The gray shading shows RS-Momentum below 100 for four of six weeks in February-March.

Even though the indicator briefly popped above 100, this pop did not last long and quickly moved back below 100. This was a sign that momentum was turning negative for RS-Ratio, and RS-Ratio ultimately crossed below 100 in the second half of March.



The blue shading shows RS-Momentum above 100 from mid-April until late May. RS-Ratio bottomed as RS-Momentum moved above 100 but did not cross above 100 until the end of May. Additionally, note that the cross above 100 in RS-Ratio came just before the early June surge in XBI.

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